

NONPARAMETRIC ESTIMATION OF THE DERIVATIVE OF AN UNKNOWN DISTRIBUTION FUNCTION IN SOME UNKNOWN TRUNCATION POINT

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SUMMARY

A nonparametric estimator is proposed for the derivative of an unknown distribution function in some unknown truncation point. It is shown to be strongly consistent and its limiting distribution is also derived.

1. Introduction

Let $\{X_n, n \geq 1\}$ be independent and identically distributed (i.i.d.) random variables with unknown distribution function $F_\theta(x)$. Suppose $F_\theta(x)$ has a finite left truncation point θ , that is $F_\theta(\theta) = 0$. For some $\epsilon > 0$ suppose that in the interval $I_\epsilon = (\theta, \theta + \epsilon)$ $F_\theta(x)$ is twice differentiable with $F''_\theta(x)$ bounded for all $x \in I_\epsilon$. Let $f_\theta(\theta) \equiv \lim_{x \downarrow \theta} F'_\theta(x) > 0$.

The problem under consideration is to estimate $f_\theta(\theta)$. Under the above conditions it is known (Galambos (1978) p. 56) that the sample minimum of the first n observations has an asymptotic exponential distribution with variance $1/n^2 f_\theta^2(\theta)$. If the sample size is large, an approximate confidence interval for θ can therefore be constructed by using the proposed estimator of $f_\theta(\theta)$. Another application is to obtain a sequential fixed width confidence interval for θ (Swanepoel and Van Wyk (1981)).

Intuitively speaking $f_\theta(\theta)$ can be estimated by the amount of "local concentration" of the observations in the neighbourhood of θ . Depending on how this is measured various estimators can be obtained. The estimator considered here is the following: Let $X_1, X_2, \dots, X_n$ be a random sample from $F_\theta(x)$ with corresponding order statistics $X_{n,1} \leq X_{n,2} \leq \dots \leq X_{n,n}$. Define

$$(1.1) \quad \hat{f}_n = (r_n - 1) \{(n+1)(X_{n,r_n} - X_{n,1})\}^{-1}$$

where $\{r_n, n \geq 1\}$ is a sequence of integers such that $1 \leq r_n \leq n$ and $r_n \rightarrow \infty$ as $n \rightarrow \infty$.

Under certain conditions $\hat{f}_n$ is strongly consistent and asymptotically normally distributed. More specifically, we obtain the following theorems:

Theorem 1

If $r_n/n \downarrow 0$ and $r_n^{-1} \log n = o(1)$ as $n \rightarrow \infty$, then $\hat{f}_n$ is a strongly consistent estimator of $f_\theta(\theta)$, that is

$$\hat{f}_n \rightarrow f_\theta(\theta) \text{ a.s. as } n \rightarrow \infty$$

Theorem 2

If $r_n = o(n^{2/3})$ as $n \rightarrow \infty$ then

$$r_n^{1/2} (f_\theta(\theta)/\hat{f}_n - 1) \xrightarrow{D} Z$$

where Z is a standard normal random variable and $\xrightarrow{D}$ denotes convergence in distribution.

The proofs of Theorem 1 and Theorem 2 will be given in section 2 and section 3 respectively. In section 4 the sequence $\{r_n, n \geq 1\}$ with $r_n = n^\nu$, $0 < \nu < 1$, is considered and a value of ν is proposed which is approximately optimal.

2. Proof of Theorem 1

For the proof of Theorem 1 we use the following results, which are stated without proof:

Let $F_\theta(x)$ satisfy the conditions in section 1. If $r_n^{-1} \log n = o(1)$ and $r_n = o(n)$ as $n \rightarrow \infty$ then with probability one

$$(2.1) \quad X_n, r_n = x_n + (r_n - n\hat{F}_n(x_n))/nF'_\theta(x_n) + O(n^{-1}r_n^{1/4}(\log n)^{3/4}) \quad \text{as } n \rightarrow \infty$$

where $\hat{F}_n(x) = n^{-1} \sum_{i=1}^n I(X_i \leq x)$ and for n sufficiently large x_n is defined by $F_\theta(x_n) = r_n/n$. Also, $I(A)$ denotes the indicator function of the event A . (See Watts (1980) p. 282).

By application of Theorem 1 and remark 2.5 of Robbins and Siegmund (1972) a law of the iterated logarithm for the sample minimum follows, that is

$$(2.2) \quad \limsup_{n \rightarrow \infty} n(X_n - 1 - \theta)f_\theta(\theta)/\log_2 n = 1 \text{ a.s.}$$

Let $\{U_n, n \geq 1\}$ be a sequence of i.i.d. random variables uniformly distributed

over $[0, 1]$ and set $T_n(u) = \sum_{i=1}^n I(U_i \leq u)$. If $\pi_n \downarrow 0$ and $n\pi_n/\log_2 n \rightarrow \infty$ and $n \rightarrow \infty$ then

$$(2.3) \quad \limsup_{n \rightarrow \infty} |T_n(\pi_n) - n\pi_n| / (2n\pi_n \log_2 n)^{1/2} = 1 \text{ a.s.}$$

(Kiefer (1972) p. 237).

Proof of the theorem

Let $G_\theta(x)$ denote the inverse function of $F_\theta(x)$, that is $G_\theta(x) = F_\theta^{-1}(x)$. By application of (2.1) and (2.2) and by expanding $x_n = G_\theta(r_n/n)$ in a Taylor series about zero we obtain

$$\begin{aligned} 1/\hat{f}_n &= (r_n - 1)^{-1} (n+1) (X_n, r_n - X_n, 1) \\ &= (r_n - 1)^{-1} (n+1) \{x_n + (r_n - n\hat{F}_n(x_n))/nF'_\theta(x_n) + \\ &\quad O(n^{-1} r_n^{1/4} (\log n)^{3/4}) - \theta + O(n^{-1} \log_2 n)\} \text{ a.s. as } n \rightarrow \infty \\ &= (r_n - 1)^{-1} (n+1) \{r_n/nf_\theta(\theta) + G''_\theta(\phi_n) r_n^2/2n^2 + \\ &\quad (r_n - n\hat{F}_n(x_n))/nF'_\theta(x_n)\} + o(1) \text{ a.s. as } n \rightarrow \infty \end{aligned}$$

for some $0 < \phi_n < r_n/n$.

Therefore,

$$(2.4) \quad 1/\hat{f}_n = 1/f_\theta(\theta) + (n+1)(r_n - n\hat{F}_n(x_n))/n(r_n - 1)F'_\theta(x_n) + o(1) \text{ a.s. as } n \rightarrow \infty$$

since $\phi_n \rightarrow 0$ as $n \rightarrow \infty$ and $G''_\theta(\phi_n)$ is bounded for n sufficiently large.

Using the definitions of $\hat{F}_n(x)$, $T_n(u)$ and x_n it follows that $n\hat{F}_n(x_n) = T_n(r_n/n)$, and applying (2.3) with $\pi_n = r_n/n$, we get

$$(2.5) \quad n\hat{F}_n(x_n) = r_n + O((r_n \log_2 n)^{1/2}) \text{ a.s. as } n \rightarrow \infty.$$

Note that $x_n \rightarrow \theta$ and hence $F'_\theta(x_n) \rightarrow f_\theta(\theta)$ as $n \rightarrow \infty$.

The proof of the theorem now follows from (2.4) and (2.5).

3. Proof of Theorem 2

It is sufficient to prove that $(r_n - 1)^{1/2} (f_\theta(\theta)/\hat{f}_n - 1) \xrightarrow{D} Z$ as $n \rightarrow \infty$. Let $U_{n,1} \leq U_{n,2}$

$\leq \dots \leq U_{n,n}$ denote the order statistics of $U_1, U_2, \dots, U_n$. It is known (Breiman (1968) p. 285) that

$$(U_{n,1}, U_{n,2}, \dots, U_{n,n}) \stackrel{D}{=} (S_1/S_{n+1}, S_2/S_{n+1}, \dots, S_n/S_{n+1})$$

for all n , where $S_i = \sum_{j=1}^i Y_j$ and the Y_j 's are independent random variables with density function $e^{-y}I(y \geq 0)$. Using this and the probability integral transformation we get

$$(3.1) \quad \hat{f}_n \stackrel{D}{=} (r_n - 1) \{ (n+1)(G_\theta(S_{r_n}/S_{n+1}) - G_\theta(S_1/S_{n+1})) \}^{-1}$$

It follows directly, by expanding $G_\theta(S_{r_n}/S_{n+1})$ and $G_\theta(S_1/S_{n+1})$ in a Taylor series about zero, that

$$(3.2) \quad (r_n - 1)^{1/2} (f_\theta(\theta) / \hat{f}_n - 1) \stackrel{D}{=} ((n+1)/S_{n+1})(r_n - 1)^{-1/2} (\sum_{j=2}^{r_n} Y_j - (r_n - 1)) \\ + (r_n - 1)^{1/2} ((n+1)/S_{n+1} - 1) + \frac{1}{2} G''_\theta(\phi_n) f_\theta(\theta) (n+1)(r_n - 1)^{-1/2} (S_{r_n}/S_{n+1})^2 \\ - \frac{1}{2} G''_\theta(\Psi_n) f_\theta(\theta) (n+1)(r_n - 1)^{-1/2} (S_1/S_{n+1})^2$$

for some $0 < \phi_n < S_{r_n}/S_{n+1}$ and $0 < \Psi_n < S_1/S_{n+1}$.

Note that by the strong law of large numbers we have

$$(3.3) \quad S_{n+1}/(n+1) \rightarrow 1 \text{ a.s. as } n \rightarrow \infty$$

and using the central limit theorem we obtain

$$(3.4) \quad (r_n - 1)^{-1/2} (\sum_{j=2}^{r_n} Y_j - (r_n - 1)) \stackrel{D}{\rightarrow} Z \text{ as } n \rightarrow \infty$$

From (3.3), (3.4) and Slutsky's theorem we get

$$(3.5) \quad ((n+1)/S_{n+1})(r_n - 1)^{-1/2} (\sum_{j=2}^{r_n} Y_j - (r_n - 1)) \stackrel{D}{\rightarrow} Z \text{ as } n \rightarrow \infty$$

By the law of the iterated logarithm for sums of i.i.d. random variables we have

$$(3.6) \quad (n+1 - S_{n+1}) = O((n \log_2 n)^{1/2}) \text{ a.s. as } n \rightarrow \infty.$$

Hence by (3.3) and (3.6) we obtain

$$(3.7) \quad (r_n - 1)^{1/2}((n+1)/S_{n+1} - 1) = O(n^{-1}r_n \log_2 n)^{1/2} \text{ a.s. } n \rightarrow \infty$$

$$= o(1) \text{ a.s. as } n \rightarrow \infty$$

since we assume $r_n = o(n^{2/3})$.

Also, $\phi_n \rightarrow 0$ and $\Psi_n \rightarrow 0$ a.s. as $n \rightarrow \infty$, so that $G''_{\theta}(\phi_n)$ and $G''_{\theta}(\Psi_n)$ are bounded for all n sufficiently large. This together with (3.3) gives

$$(3.8) \quad \frac{1}{2} G''_{\theta}(\phi_n) f_{\theta}(\theta) (n+1) (r_n - 1)^{-1/2} (S_{r_n}/S_{n+1})^2 = o(1) \text{ a.s. as } n \rightarrow \infty$$

Similarly, we have

$$(3.9) \quad \frac{1}{2} G''_{\theta}(\Psi_n) f_{\theta}(\theta) (n+1) (r_n - 1)^{-1/2} (S_1/S_{n+1})^2 = o(1) \text{ a.s. as } n \rightarrow \infty.$$

The proof of Theorem 2 follows from (3.2), (3.5), (3.7), (3.8) and (3.9).

4. Optimal Choice of r_n

In this section we consider sequences $\{r_n, n \geq 1\}$ of the form $r_n \sim n^{\nu}$ for $0 < \nu < 1$. Define the asymptotic mean square error

$$\begin{aligned} \text{AMSE} &= E(1/\hat{f}_n - 1/f_{\theta}(\theta))^2 \\ &= \text{var}(1/\hat{f}_n) + E^2(1/\hat{f}_n - 1/f_{\theta}(\theta)). \end{aligned}$$

Proceeding heuristically, as in Bloch and Gastwirth (1968), we have from Theorem 2 that

$$(4.1) \quad \text{var}(1/\hat{f}_n) \approx 1/r_n \hat{f}_{\theta}^2(\theta).$$

Also, for n large we have

$$1/\hat{f}_n \approx (r_n - 1)^{-1} (n+1) \{ (U_{n, r_n} - U_{n, 1})/f_{\theta}(\theta) + \frac{1}{2} G''_{\theta}(0) (U_{n, r_n}^2 - U_{n, 1}^2) \}$$

and by noting that $U_{n, i}$ has a beta distribution with parameters i and $n-i+1$ it follows that

$$(4.2) \quad E(1/\hat{f}_n - 1/f_{\theta}(\theta)) \approx r_n G''_{\theta}(0)/2n.$$

From (4.1) and (4.2) we have for large n that

$$(4.3) \quad \text{AMSE} \approx 1/n^\nu f_\theta^2(\theta) + c^2 n^{2\nu-2}$$

with $c = \frac{1}{2} G_\theta''(0)$.

It is easily seen from (4.3) that the optimal choice for ν is

$$(4.4) \quad \nu = 2/3 - \log(2c^2 f_\theta^2(\theta))/3 \log n \approx 2/3, \text{ for } n \text{ large.}$$

In Theorem 2 we assume $r_n = o(n^{2/3})$ as $n \rightarrow \infty$ which implies that ν should be less than $2/3$. This together with (4.4) therefore suggests that a value of ν just less than $2/3$ should be taken.

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